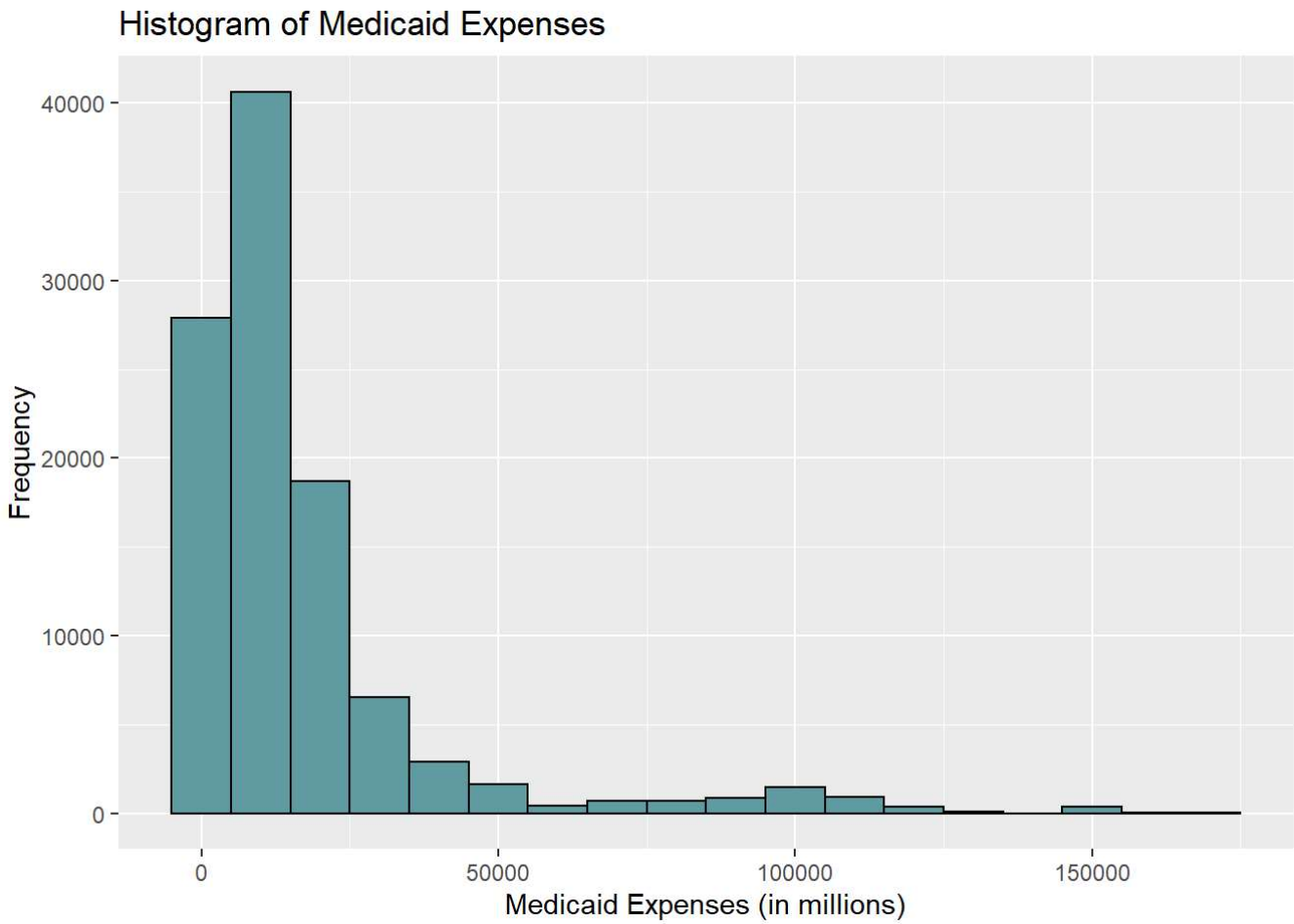
# Chronic_Cancer_Analysis

2025-07-03

## Medicaid Expenses Distribution

```
ggplot(final_df, aes(x = Medicaid_Expenses)) +
  geom_histogram(binwidth = 10000, fill = "cadetblue", color = "black") +
  labs(title = "Histogram of Medicaid Expenses", x = "Medicaid Expenses (in millions)", y = "Fre
quency")
```

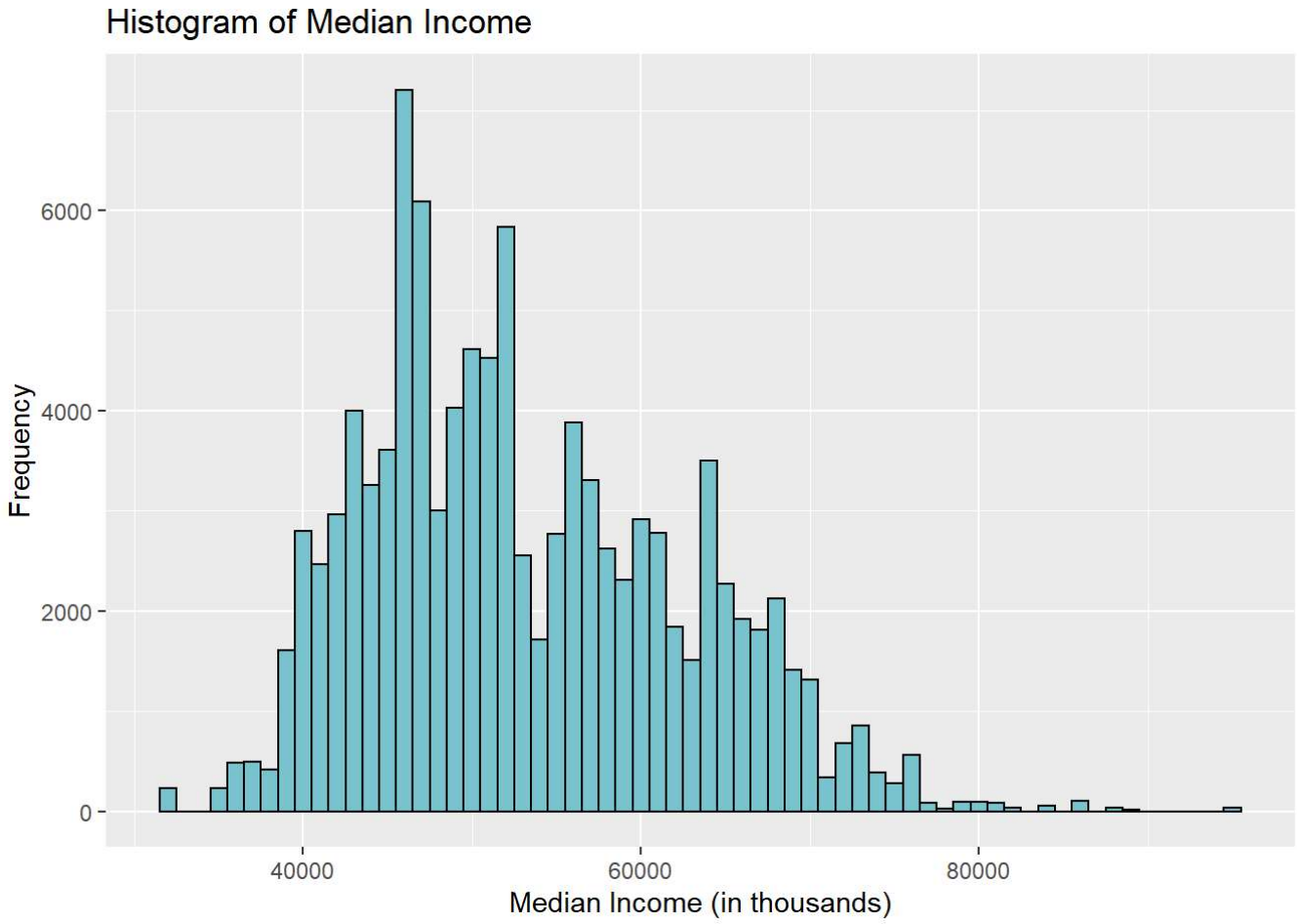



## Insights

1. Right-skewed distribution This means: Most Medicaid expenses are relatively low. A small number of observations have extremely high expenses, creating a long tail.
2. Majority of States/Populations Fall Below ~$25,000M The mode (most frequent range) of Medicaid expenses is likely in the $0–$25,000 million range. Suggests that many states or regions spend moderately, while only a few states have very high Medicaid costs. —

# Median Income Distribution

```
ggplot(final_df, aes(x = Median_Income)) +
  geom_histogram(binwidth = 1000, fill = "cadetblue3", color = "black") +
  labs(title = "Histogram of Median Income", x = "Median Income (in thousands)", y = "Frequenc
y")
```




# Insights

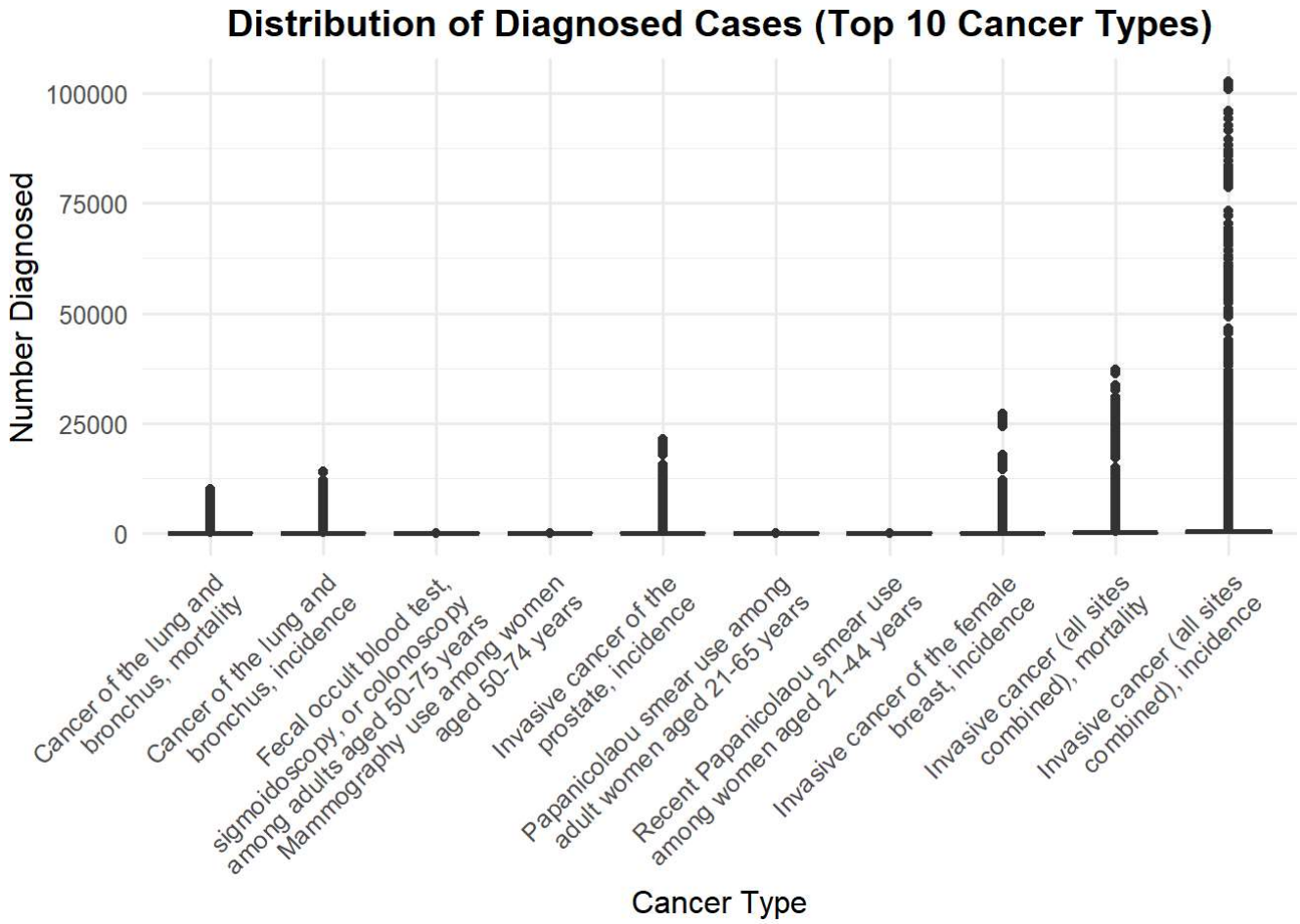
1. Skewness of Distribution: The histogram is right-skewed (positively skewed), meaning most of the data is concentrated on the lower income range (e.g., $40k–$60k), with a long tail toward the higher end. This suggests fewer high-income observations, and many moderate- or low-income ones.
2. Mode and Central Tendency: The peak (mode) appears to be around $45,000 to $50,000. This is where the most frequent income values lie. —

# Diagnosed Cases Distribution (Top 10 Cancer Types)

```
top_types <- final_df %>%
  group_by(Type) %>%
  summarise(median_cases = median(Number_Diagnosed, na.rm = TRUE)) %>%
  arrange(desc(median_cases)) %>%
  slice_head(n = 10) %>%
  pull(Type)

filtered_df <- final_df %>% filter(Type %in% top_types)
filtered_df$Type <- str_wrap(filtered_df$Type, width = 30)
filtered_df$Type <- reorder(filtered_df$Type, filtered_df$Number_Diagnosed, FUN = median)

ggplot(filtered_df, aes(x = Type, y = Number_Diagnosed)) +
  geom_boxplot(fill = "lightcoral") +
  theme_minimal(base_size = 12) +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    plot.title = element_text(face = "bold", hjust = 0.5)
  ) +
  labs(
    title = "Distribution of Diagnosed Cases (Top 10 Cancer Types)",
    x = "Cancer Type",
    y = "Number Diagnosed"
  )
```

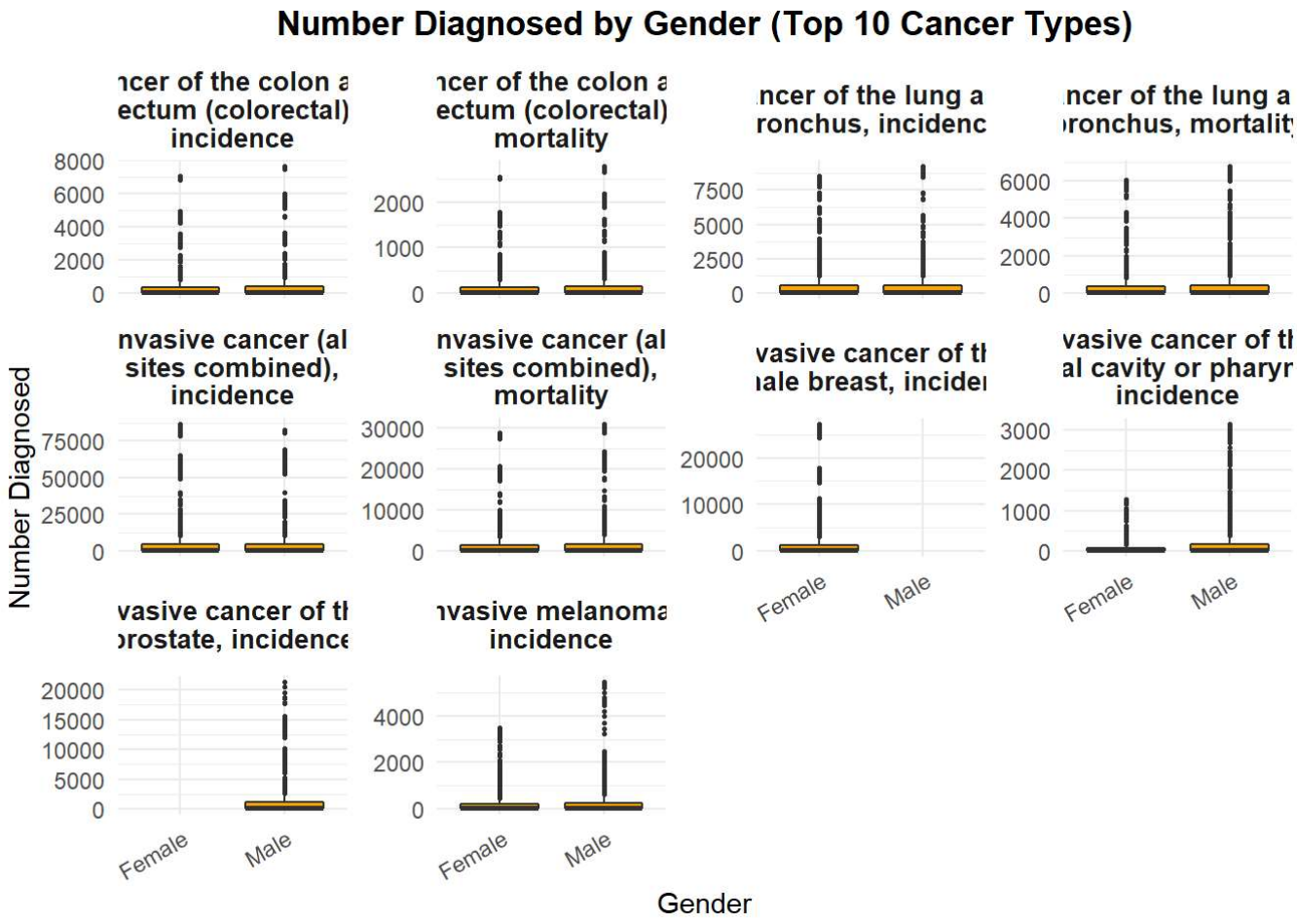


# Diagnosed Cases by Gender

```
gender <- final_df %>% filter(StratificationCategory1 == "Gender")
top_types <- gender %>%
  group_by(Type) %>%
  summarise(Total = sum(Number_Diagnosed, na.rm = TRUE)) %>%
  arrange(desc(Total)) %>%
  slice_head(n = 10) %>%
  pull(Type)

filtered_gender <- gender %>% filter(Type %in% top_types)
filtered_gender$Type <- str_wrap(filtered_gender$Type, width = 25)

ggplot(filtered_gender, aes(x = Stratification1, y = Number_Diagnosed)) +
  geom_boxplot(fill = 'orange', outlier.size = 0.5) +
  facet_wrap(~Type, scales = 'free_y') +
  labs(
    title = 'Number Diagnosed by Gender (Top 10 Cancer Types)',
    x = 'Gender',
    y = 'Number Diagnosed'
  ) +
  theme_minimal(base_size = 11) +
  theme(
    strip.text = element_text(face = "bold", size = 10),
    axis.text.x = element_text(angle = 30, hjust = 1),
    plot.title = element_text(hjust = 0.5, face = "bold")
  )
```

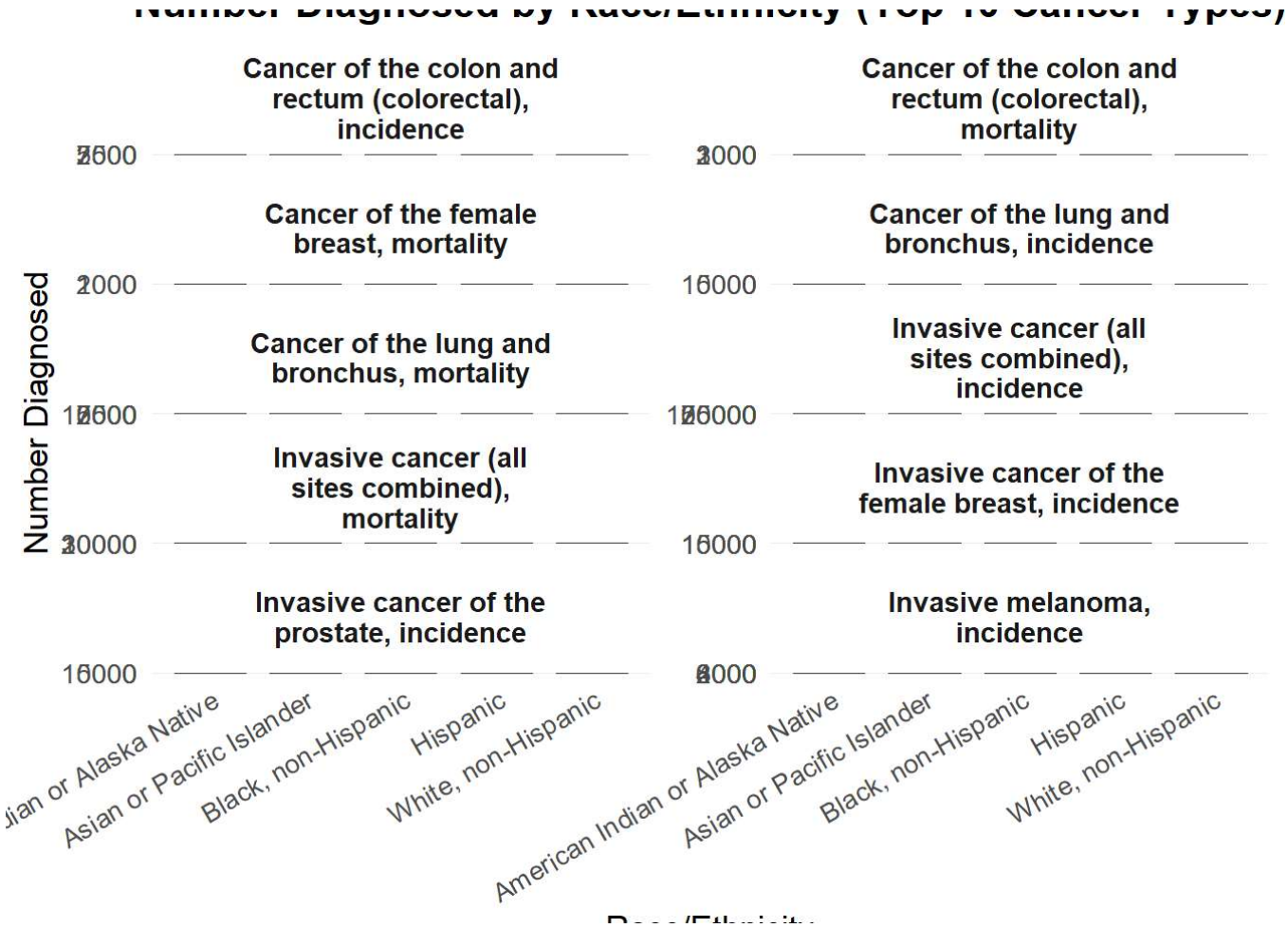

Number Diagnosed by Gender (Top 10 Cancer Types)
ncer of the colon a
ectum (colorectal)
incidence
ncer of the colon a
ectum (colorectal)
mortality
ncer of the lung a
ronchus, incidenc
ncer of the lung a
ronchus, mortalit
nvasive cancer (al
sites combined),
incidence
nvasive cancer (al
sites combined),
mortality
vasive cancer of tl
ale breast, incide
vasive cancer of tl
al cavity or pharyr
incidence
vasive cancer of tl
rostate, incidence
nvasive melanoma
incidence
Number Diagnosed
Female
Male
Gender

# Diagnosed Cases by Race/Ethnicity

```
race <- final_df %>% filter(StratificationCategory1 == 'Race/Ethnicity')
top_types <- race %>%
  group_by(Type) %>%
  summarise(Total = sum(Number_Diagnosed, na.rm = TRUE)) %>%
  arrange(desc(Total)) %>%
  slice_head(n = 10) %>%
  pull(Type)

filtered_race <- race %>%
  filter(Type %in% top_types) %>%
  mutate(Type = str_wrap(Type, width = 25))

ggplot(filtered_race, aes(x = Stratification1, y = Number_Diagnosed, fill = Stratification1)) +
  geom_boxplot(outlier.size = 0.5) +
  facet_wrap(~Type, scales = "free_y", ncol = 2) +
  labs(
    title = "Number Diagnosed by Race/Ethnicity (Top 10 Cancer Types)",
    x = "Race/Ethnicity",
    y = "Number Diagnosed"
  ) +
  theme_minimal(base_size = 13) +
  theme(
    strip.text = element_text(size = 11, face = "bold"),
    axis.text.x = element_text(angle = 30, hjust = 1),
    legend.position = "none",
    plot.title = element_text(hjust = 0.5, face = "bold")
  )
```



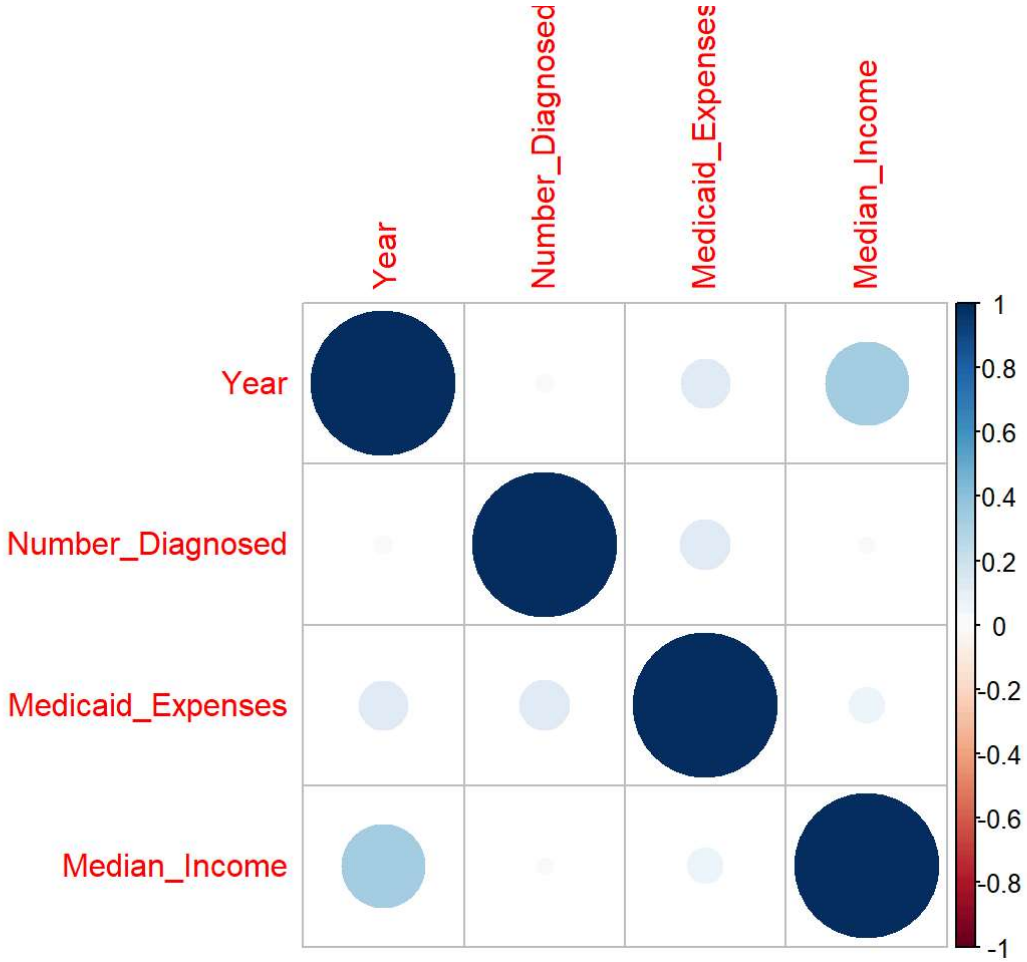

# Insights

1. Non-Hispanic White and Non-Hispanic Black populations generally show higher median counts, suggesting higher reported diagnoses.
2. Lung and Bronchus Cancer (Incidence & Mortality): Mortality and incidence are both notably high for Non-Hispanic Black and White groups.
3. Female Breast Cancer (Incidence & Mortality): High incidence among Non-Hispanic White and Black women. However, mortality appears disproportionally higher for Non-Hispanic Black women, suggesting disparities in treatment access or diagnosis timing.
4. Prostate Cancer (Incidence): Extremely high median diagnoses for Non-Hispanic Black males compared to other races. —

# Correlation Matrix

```
cor_matrix <- final_df %>%
  select(where(is.numeric)) %>%
  cor(use = "pairwise.complete.obs")

corrplot::corrplot(cor_matrix, method = "circle")
```




# Insights

Strong Positive Correlations:

1. Number_Diagnosed vs. Medicaid_Expenses: strong positive correlation (dark blue, close to 1) indicates that as the number of diagnosed cases increases, Medicaid expenses also rise. Suggests that disease burden is a key driver of healthcare costs.
2. Medicaid_Expenses vs. Year: moderate to strong positive correlation suggests Medicaid expenses are increasing over time, possibly due to rising treatment costs or broader coverage.
3. Median_Income vs. Year: positive correlation indicates median income has increased over the years (likely due to inflation and economic growth).

Weak/Negligible Correlations:

1. Number_Diagnosed vs. Median_Income: most no correlation — indicates that income levels are not strongly related to the number of diagnosed cancer cases in the data.
2. Year vs. Number_Diagnosed: very weak — implies the number of diagnoses may not be significantly increasing or decreasing over time (could be stable or vary by type). —